

# A moratorium on Ai data centres – The Council of Canadians

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## Source Content

RE: A moratorium must be placed on the development of data centres and their associated infrastructure until the full impacts are known to the public. Dear Ministers, Premier, and Impact Assessment Agency, The environmental, climate, and community organizations undersigned are writing to request that a moratorium be placed on the development of Hyperscale Artificial Intelligence (AI) data centres and their associated infrastructure until the full impacts are known to the public. At present, these are our four major concerns: - Power demands from proposed data centres are significant, which may strain the grid, compete for supply, and deepen fossil fuel dependence in the province. - Natural gas production and consumption will markedly increase to generate the required power, which has negative implications for the environment, public health, and climate change, as well as real-world costs. - Data centres, natural gas power plants, and any carbon capture and storage technology used onsite all require large volumes of water, which many watersheds in Alberta cannot afford to lose. - The economic benefits for the public aren't clear, engagement has been limited, and processes and projects are not transparent. We are concerned that Alberta and Canada have allowed multiple data centres (Mihta Askiy, Synapse, and Wonder Valley) to bypass the Environmental Impact Assessment (EIA) process. The purpose of an EIA, in the provincial government's own words, is to allow "companies and government decision makers to examine the effects that the proposed project may have on the environment and determine if the project is of public interest." By bypassing this due process, government officials have decided on everyone's behalf that hyperscale AI data centres are in the public interest without any substantiating evidence. However, research on existing data centres has found that their operations can have significant negative impacts on the environment and public. Data centres have been found to reduce air and water quality, increase heat exposure and carbon emissions, cause noise pollution, and result in the loss of public lands, natural spaces, and agricultural areas. The public deserves to know the full scope and implications of these impacts. Until this occurs, the public, and our shared watersheds, landscapes, and atmospheres must be protected by a moratorium on hyperscale AI data centre development. Proposed data centres are already requesting two times more electricity the entire province of Alberta regularly uses. The computing power required by AI means massive energy demands. For example, operating Synapse Data Centre Inc., proposed in the town of Olds, would require 1,400 megawatts (MW) of power. This is equal to the entire city of Edmonton's power demand, and is just one of dozens of similarly energy-intensive data centres hoping to operate in the province. The Alberta Electric Systems Operator (AESO) reports that requests from data centres to connect to the grid have "surged" since 2024 and "continue to arrive, with over 20,000 MW of data centre load now on the

AESO project list” (Figure 1). (source: March Update on Data Centres, September Update on Data Centres). To put this immense demand in perspective, the entire province of Alberta is generating and consuming around 10,000 MW as this is being written. Based on AESO’s last update in September of 2025, cumulative demand for power from data centres has increased 9,900% percent since the beginning of 2024, with load well outpacing generation. AESO states that most data centres are “not contributing sufficient or timely generation, creating grid reliability issues” (Figure 1). However, some projects are still pushing ahead and will be allowed to connect to the grid before their accompanying power plants are finished construction. Announced in early July of 2026, Meta’s 970 MW data centre in Sturgeon County plans to start operating one to two years before construction of the Greenlight Electricity Centre, its neighbouring natural gas-fired power plant, is complete. As a pre-emptive measure, AESO has capped supply for data centres at 1,200 MW and currently requires new operators to provide their own power rather than draw from the shared grid. Many proposed projects indicate intentions to construct large natural gas-based generators, which will increase air pollutants and greenhouse gas emissions, as well as compete with the public and other industries for the supply of natural gas within the province. In other jurisdictions, this competition has ultimately raised energy costs for the public. Data centres will be powered by natural gas, which has significant environmental, health, and climate impacts. The chemical compounds released when burning natural gas (largely comprised of methane) decrease air quality and are associated with increased short- and long-term health burdens on the public, including respiratory illnesses, cardiovascular diseases, neurological disorders, poor reproductive outcomes, and premature mortality. Planned in the Municipal District of Greenview and purported to be the ‘largest data centre on Earth,’ Wonder Valley would need a 7,500-MW power plant to operate and require “about 10% of all gas supply in Alberta once fully operational.” It would emit over 25 million tonnes of greenhouse gases alone, reversing all the progress the province has made on emissions reductions since 2015 by phasing coal power out of the provincial grid. This backsliding has real costs, as intensifying carbon emissions contributes to climate change, and already we are seeing the cascading effects that decades of climate change denial, dithering, and inaction have wrought. Alberta is located in one of the world’s four “mega-drying regions,” where terrestrial water storage is being lost at increasingly faster rates due to climate change. The landscape is losing its natural ability to retain water, which has profound implications for our livelihoods. A drier Alberta means increased water insecurity — less water reliably available for the economy and to meet all our society’s essential needs (think health, hygiene, sanitation, safety, and emergency response). It means increased costs for food, as the loss of crop and livestock production squeezes supply and creates shortages, as well as higher energy costs, with extreme temperatures becoming more common. In this rapidly changing climate, droughts, wildfires, and floods all become more frequent, severe, and unpredictable, and with that comes increased expenses for responding to, managing, mitigating, and preventing the next natural disaster. Data centres and their power plants require large amounts of water “Water risk near data centre locations also varies significantly across provinces, with locations in Alberta and Saskatchewan displaying especially high rates of water risk”. Carlo, A., & Rolheiser, L., 2026 AI data centres are known to consume considerable volumes of water for both cooling purposes and when generating electricity, which can contribute to local water scarcity, particularly in drought-prone regions. In addition to quantity, water quality can also be impacted, with research finding that the evaporative cooling processes often employed at data centres produce wastewater with high concentrations of contaminants. On the same day the Alberta government and Meta announced the development of a 970-MW data centre in Sturgeon County, it was reported Meta “flushed bacteria-contaminated water into public sewers during construction” of another data centre in Cheyenne, Wyoming. Many of the data centre developers in Alberta assert they will require limited water withdrawals or use ‘closed loop systems,’ but exact volumes, technologies, and management procedures remain vague. An EIA would clarify the total water use and better inform decision-makers on the sustainability of this industry. There also is seemingly little consideration for the increased water use and consumption that results from

powering the data centres. Increasing natural gas supply requires more hydraulic fracturing, aka ‘fracking,’ which injects water deep inside the earth, effectively taking it out of the provincial water balance for foreseeable generations. Fracking can also induce seismic activity and contaminate groundwater supplies. Assuming new combined cycle natural gas-fired turbines are constructed (as has been indicated in Meta’s and Synapse’s projects), and depending on the input fuel and cooling technology used, powering hyperscale AI data centres could consume anywhere between 833,000 (conventional gas, dry cooling) to 5.6 million (shale gas, cooling pond) litres of water annually per MW over their life cycle. This means that each year, a 1,000-MW plant would consume 833 million to 5.6 billion litres, the same amount of water as the annual use of 12,000 to 79,000 Albertans. If all 20,000-plus MW requested by currently proposed data centres came to fruition, their power plants alone would consume as much water as 234,000 to 1.6 million Albertans each year. Water use intensity further increases if the power plants use carbon capture and storage (CCS) technologies to mitigate the climate-warming emissions their operations release. Research has found that operational water consumption increases by 29 to 81 percent at combined cycle power plants with CCS, depending on cooling type. Adding all this new water demand is especially worrisome considering many watersheds, particularly in dry Southern Alberta, are already overallocated and intensely used; to continue to reliably provide to all those reliant on them, rivers like the Oldman and the Bow need more water flowing through them, not less. The economic benefits for the public aren’t clear “The concentration of future development in Alberta raises a question that extends beyond the province’s borders. Alberta’s population bears the costs of this expansion – through electricity demand, emissions, and local resource use – yet the economic returns on the underlying assets may accrue elsewhere: to investors and firms located outside the province, or outside Canada altogether.” The Energy Mix, 2025 Researchers at York University have noted that Alberta is among one of the few jurisdictions in North America without a sales or property tax on machinery and equipment, which is one of the main ways governments make money on data centres. Although the Government of Alberta recently put a levy of 2 percent on the computing equipment at hyperscale data centres connected to the grid, it is “fully deductible against provincial corporate income taxes, imposing no additional burden once a facility becomes profitable.” Projects that generate their own power have even lower rates. Tax breaks are also available under the Municipal Government Act, and if offered, could allow hyperscale data centres to be exempt or defer paying municipal taxes for up to 15 years. The researchers observed that all these incentives for “incoming facilities place significant demands on provincial infrastructure and services while contributing little to provincial revenues.” As no comprehensive EIAs have been undertaken, the full cost-benefits of these projects are unknown. What economic impacts will be incurred on Alberta’s river basins, as diverting and consuming large volumes of water to cool data centres and generate power negatively impacts their function and integrity? What health and climate costs will there be from increasing extraction and consumption of natural gas, and releasing even greater amounts of carbon emissions into the atmosphere? How much will mitigation cost? What is the cost to Albertan’s health, communities, and environment if mitigation is ineffective, or insufficient? What is the cost to Canada if we continue on as climate laggards, deepening our dependence on fossil fuels? The public deserves to know the impact of AI data centres. Other environmental issues associated with AI data centres include noise pollution, increased heat exposure, and loss or alteration of natural areas and agricultural lands and their associated benefits. At a time when we are already dealing with energy uncertainty, water shortages, concurrent natural disasters, climate change, ecosystem loss, and increasing health risks, AI operators need to prove that the benefits and opportunities outweigh the extensive costs. AI data centres have real-world implications for nearby communities, but as projects are pushed through rapidly without EIAs, whether those impacts are acceptable or sustainable remains unknown. EIAs also investigate the social, economic, and health impacts, providing a cost-benefit analysis to better weigh a given project. In the absence of EIAs, the public is forced to take companies at their word, which is currently limited. If AI data centres are as beneficial, sustainable, and lucrative as their proponents would claim, then they have nothing to fear from a

robust EIA. The process would serve to increase transparency, proactively identify issues and address public concerns, and foster local goodwill. But, for reasons lacking explanation, the impact assessment process is being bypassed altogether, which places us all at undue risk. A moratorium on hyperscale data centre development must be enacted, and environmental impact assessments must be mandatory. We request that a moratorium be placed on hyperscale data centre development, inclusive of their power plants and associated infrastructure, until robust public engagement can be undertaken and the full scale and scope of impacts are understood. Precedent already exists in Alberta, where moratoriums were placed on coal and renewable energy development to undertake public engagement on the industries. Additionally, EIAs must be mandatory for all hyperscale data centres and accessory infrastructure. We look forward to your confirmation of receipt and response to our request and concerns. Sincerely, Alberta Wilderness Association  
Calgary Climate Hub   Climate Justice Edmonton   Calgary Philippines Solidarity Organizing Group   The Council of Canadians Edmonton Chapter   Southern Alberta Group for the Environment   Water Watchers and The Council of Canadians